Processing subject negative concord in three American Englishes

American Englishes vary in the types of negative sentences they permit. Negative concord, in which multiple syntactic negations contribute to a single semantic negation (*I didn't bake no cookies*, 'I baked no cookies'), is rare in Mainstream US English but common in many varieties, including Appalachian and African-American Englishes (Labov, 1972; Wolfram & Fasold, 1974). Even among varieties that permit negative concord, the frequency of constructions varies: Appalachian English (AppE) and African-American English (AAE) both permit two patterns of subject negative concord, but corpus data suggest preverbal subject negative concord (*nobody didn't help,* 'nobody helped') is favored in AppE, and postverbal (*didn't nobody help*) in AAE.

Prior work has explored the acceptability and interpretation of sentences with two syntactic negations in Mainstream US English (MUSE) (Blanchette & Lukyanenko, 2019), but to date, little or no work has addressed how negative concord is processed by people who produce it. To compare processing across levels of use and familiarity, the current study explores processing of preverbal subject negative concord in three groups of participants: AppEnglish users in East Tennessee (n = 30), AAE users in Washington DC (n = 22) and MUSE users in a Mid-Atlantic university town (n = 50). Data collection is ongoing for the AppE and AAE groups.

Participants' eyes were tracked as they read pairs of sentences. Critical sentences contained a negative subject followed by a negative auxiliary (e.g., *Nobody didn't try the recipe*), and were preceded by context sentences that supported either a negative concord reading (NC, 'nobody tried the recipe') or a double negation reading (DN, 'everybody tried the recipe'; e.g., *Carol published her awful/famous recipe in the church newsletter*). Each trial was followed by a question probing interpretation. We expected DN readings to be accessible for all participants, and for NC readings to be accessible to AppE and AAE participants, but not MUSE participants. We analyzed interpretations and two measures of comprehension difficulty: total time spent reading the critical sentence and time spent rereading the context sentence after first fixating the critical sentence.

Reading and rereading times were comparable across contexts for AppE and AAE users, but much longer in NC than DN contexts for MUSE users (Figs1-2). This pattern was supported by a mixed-effects model which showed a reliable effect of context type, and a reliable interaction between context and a MUSE-others contrast. This is consistent with prior results from MUSE users (Blanchette & Lukyanenko, 2019), and suggests that processing may differ qualitatively depending on a structure's presence (AppE, AAE) or absence (MUSE) in a participant's variety.

Interpretations showed a somewhat different pattern: All three groups were influenced by context to a strikingly similar degree, but AAE and especially AppE users reached more NC interpretations overall than MUSE users did. A mixed-effects model showed only reliable main effects of context type and a MUSE-others contrast. Numerical differences between AppE and AAE users' responses point in the direction of the corpus frequency differences but did not reach significance in the current sample.

These results show that, despite showing a similar degree of flexibility in interpretation, MUSE users struggle to reach NC interpretations. These patterns also suggest broad similarities in the processing of preverbal subject negative concord sentences by AppE and AAE users, despite likely differences in frequency of exposure. A more fine-grained analysis of reading patterns is underway to further explore the depth of this similarity.

References

Blanchette, F. & Lukyanenko, C. (2019). Unacceptable grammars? An eye-tracking study of English Negative Concord. *Language and Cognition* 11.

Labov, W. (1972). Negative attraction and negative concord in English grammar. *Language* 48.

Wolfram, W. & Fasold, R. (1974). *The study of social dialects in American English*. Prentice Hall.

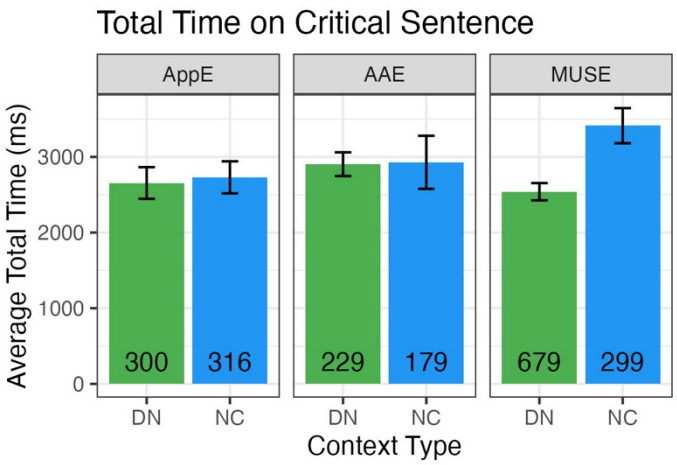

Figure 1. Total time spent reading the critical sentence, for trials in which participants' interpretation matched the context. Number of trials contributing to each value is shown on the bar. Patterns remain similar when all trials are included.

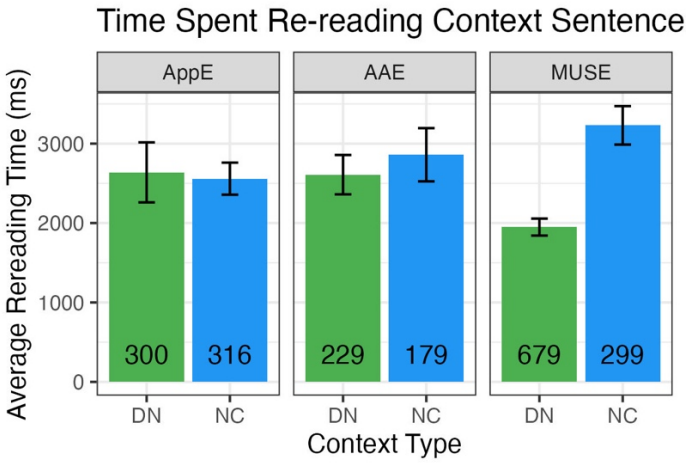

Figure 2. Total time spent rereading the context sentence, for trials in which participants' interpretation matched the context. Number of trials contributing to each value is shown on the bar. Patterns remain similar when all trials are included.

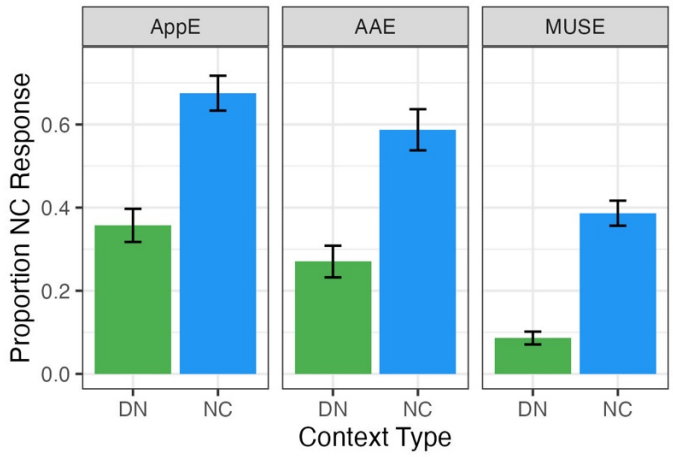

Figure 3. Proportion NC interpretations.